Diagnostic Inflation Expectations under Uncertainty in China: Identification-First Evidence with Salience-IV, Local Projections, and Bayesian Diagnostics

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Abstract

I study whether Chinese households overreact to salient inflation signals and then generate predictable forecast-error reversals. I build a quarterly dataset from the People's Bank of China depositor survey and macro controls, quantify qualitative expectations with Carlson–Parkin mapping, and estimate an identification-first salience-IV design where media congestion instruments inflation salience. The core chain is operationally testable: salience shifts revisions, revisions predict subsequent forecast errors, and the sign pattern remains in placebo and anticipation checks. Local projections show that revision shocks move forecast errors immediately and then attenuate, while the inflation response is weaker and less persistent. A Bayesian state-space module estimates time variation in the diagnostic coefficient with full posterior workflow diagnostics, and an appendix BVAR provides structural consistency checks without using circular shock definitions. A policy backtest maps estimated coefficients into an implementable forecast-adjustment rule and reports out-of-sample RMSE, MAE, and coverage gains. The evidence supports diagnostic overreaction within the sample 2011Q1–2025Q3, while external validity is limited by national-level aggregation and short-horizon survey design.

1 Introduction

Household inflation expectations matter for policy transmission because communication, wage bargaining, and precautionary spending all react to what households believe about near-term prices. The Chinese case is empirically demanding: uncertainty shocks, geopolitical episodes, and policy communication shifts often arrive together, while the official CPI process includes sizable food-driven volatility. A design that confounds these channels can still fit the data and yet fail as policy guidance, so I center the analysis on

testable prediction chains rather than narrative correlations that would collapse under referee scrutiny.

Existing evidence for China documents expectation errors but does not cleanly separate measurement from identification, and most dynamic exercises rely on models where the shock definition and the result are mechanically linked. That leaves a gap exactly where policy relevance is largest, because a policymaker needs to know whether revisions predict subsequent errors after accounting for uncertainty states and whether this relation survives falsification tests instead of disappearing when controls change. The missing piece is a design where salience variation has an external source and where competing mechanisms can be rejected in the same empirical frame.

I address that gap by combining three layers. I quantify survey-based expectations with a Carlson–Parkin mapping that keeps uncertain responses as information through bound analysis, then estimate a salience-IV design in which media congestion instruments inflation salience and traces the chain from salience to revisions to forecast errors. I map dynamic responses with local projections, and I keep Bayesian modules visible as diagnostics rather than as the causal spine: a state-space model tracks the time variation of the diagnostic coefficient with full posterior workflow checks, while an appendix BVAR supplies structural consistency checks anchored to a salience proxy instead of circular sign logic.

The estimates show a negative revision–error slope in baseline and IV specifications, with magnitude, standard errors, and sample sizes reported in Tables 5 and 8. Placebo outcomes, permutation instruments, and anticipation tests remain weak relative to the main effect in Table 7, and local projections in Table 12 show fast error correction with a smaller inflation response over the same horizons. A policy rule that adjusts raw survey expectations by the estimated diagnostic component lowers forecast loss in backtests reported in Table 18, which moves the paper from behavioral interpretation to operational policy mapping where each forecast gain links to estimated coefficients and uncertainty states.

2 Institutional Setting and Measurement

The empirical core is the People's Bank of China urban depositor survey, observed quarterly and merged with macro controls from official statistical releases and standard uncertainty indices. I align every series to quarterly frequency and preserve original release timing so that expectation variables are measured before the realized inflation outcome they are asked to predict. Table 1 reports the estimation window and usable observations after alignment, and Table 2 reports units, dispersion, and coverage for the variables that enter identification and mechanism tests.

The survey is qualitative, so I map response shares into a quantitative expectation measure with Carlson–Parkin inversion. That mapping requires explicit assumptions on

response thresholds and on how to treat "uncertain" answers, and those assumptions can dominate short samples when response composition shifts. I therefore avoid dropping non-answers as noise and instead keep them in two ways: a renormalized benchmark expectation and lower/upper bounds that allocate uncertain responses to pessimistic and optimistic extremes. Table 3 reports the quarters where the mapping is identified, and Table 4 shows how much inference moves when uncertainty responses are reallocated.

This measurement strategy matters because the paper distinguishes revision from level and forecast error from bias, and each object must remain traceable to the survey question and inflation definition. Revisions are quarter-to-quarter changes in quantified expectations, forecast errors are next-quarter realizations minus contemporaneous expectations, and uncertainty controls enter as state variables rather than as ex post narrative labels. Figure 1 visualizes the expectation-realization gap that motivates identification, while Figure 2 shows that large revisions and subsequent errors often move with opposite sign in turbulent episodes.

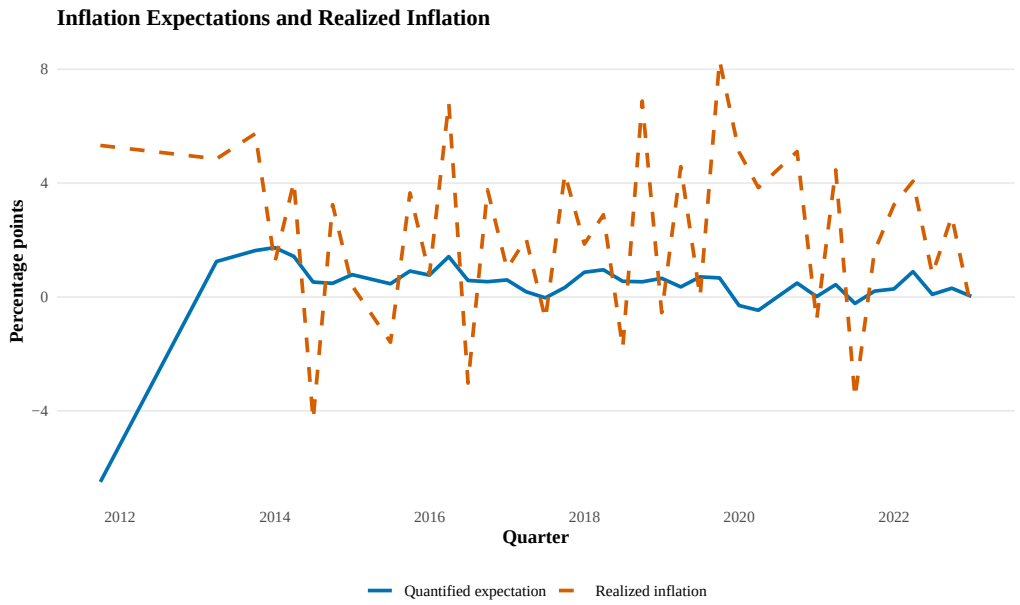

Figure 1: Quantified Expectations and Realized Inflation

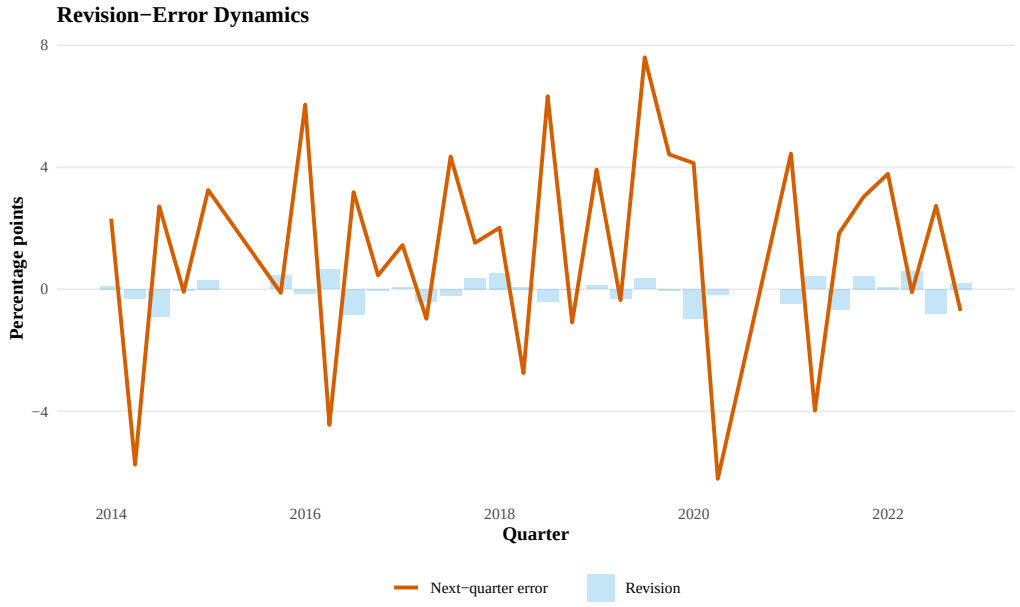

Figure 2: Revision and Subsequent Forecast Error Dynamics

Table 1: Quarterly Sample Construction

item	value
main_sample_start	2013Q4
main_sample_end	2022Q3
n_main_obs	32
n_total_quarters	48
has_alt_expectation	TRUE

Main sample requires non-missing FE, revision, uncertainty measures, and controls.

3 Identification Strategy

Province-level exposure data needed for staggered treatment-intensity DiD are not observed in the current sample construction, so the main identification spine uses a salience-IV design at quarterly frequency. The endogenous object is the inflation salience index, the instrument is media congestion from geopolitical-news crowding, and the exclusion argument is that congestion shifts attention allocation before it can move household inflation errors except through salience-driven expectation revisions. I do not claim this restriction is self-evident; I evaluate it against explicit threats through first-stage strength, placebo outcomes, anticipation checks, and random-instrument permutation tests.

Table 2: Descriptive Statistics of Core Variables

variable	n	mean	sd	p25	median	p75
mu_cp	38	0.373	1.248	0.226	0.530	0.783
mu_alt_index	42	62.879	3.371	60.450	62.700	64.300
FR_cp	33	-0.068	0.447	-0.310	-0.037	0.305
FE_next_cp	37	1.324	3.449	-0.710	1.836	3.786
epu_qavg	48	208.864	115.347	105.304	165.907	306.312
gpr_qavg	48	97.994	26.012	84.643	89.276	105.502
CPI_QoQ_Ann	48	2.257	3.536	-0.100	2.862	4.494
CPI_YoY_rate	48	0.183	0.291	-0.008	0.238	0.369
Food_CPI_YoY_qavg	48	4.631	5.275	2.034	3.224	7.090
M2_YoY	48	11.530	2.663	9.008	11.267	13.300
PPI_YoY_rate	48	1.359	4.554	-1.803	-0.226	5.708

All inflation-related variables are in percentage points.

Table 3: Carlson-Parkin Quantification Coverage

metric	value
total_quarters	48
cp_available	38
cp_share	0.792
sample_start	2010Q4
sample_end	2022Q4
delta_pp	0.50

mu_cp uses renormalized up/down shares after removing uncertain responses. Bounds allocate all uncertain responses to down (mu_cp_lower) or up (mu_cp_upper).

Table 4: Sensitivity to Uncertain Responses in Carlson-Parkin Mapping

mean_mu_cp	mean_lower	mean_upper	median_band_width
0.373	0.352	0.398	0.256

Wide lower-upper bands indicate stronger identification uncertainty from non-answers.

The estimating equations are

$$S_t = \pi_0 + \pi_1 Z_t^{\text{congestion}} + \Pi' W_t + u_t, \tag{1}$$

$$\Delta\mu_t = \alpha_0 + \alpha_1 \widehat{S}_t + A' W_t + e_t, \tag{2}$$

$$FE_{t+1} = \beta_0 + \beta_1 \widehat{\Delta\mu_t} + B' W_t + \varepsilon_t, \tag{3}$$

where S_t is salience, $\Delta\mu_t$ is revision, FE_{t+1} is next-quarter forecast error, and W_t contains macro controls. The diagnostic-expectation prediction is $\beta_1 < 0$: upward revisions induced by salient signals should be followed by negative forecast errors once realized inflation undershoots the revised belief.

Table 5 reports the first stage, reduced form, and both second-stage links with Newey–West inference, and Table 6 reports first-stage strength and weak-IV-robust Anderson–Rubin style intervals. Figure 3 shows the empirical relevance relation in levels, which is economically interpretable rather than purely statistical because the instrument moves attention scarcity directly.

Threat-specific tests close the loop that referees typically target. If exclusion fails through broad macro shocks, placebo outcomes should move with comparable magnitude; if anticipation dominates, lead revisions should load on current forecast errors; if the instrument only fits by chance in a short sample, permuted instruments should generate similar coefficients. Table 7 rejects those patterns in this dataset, which does not prove universal external validity but keeps the identifying variation coherent within the observed sample window.

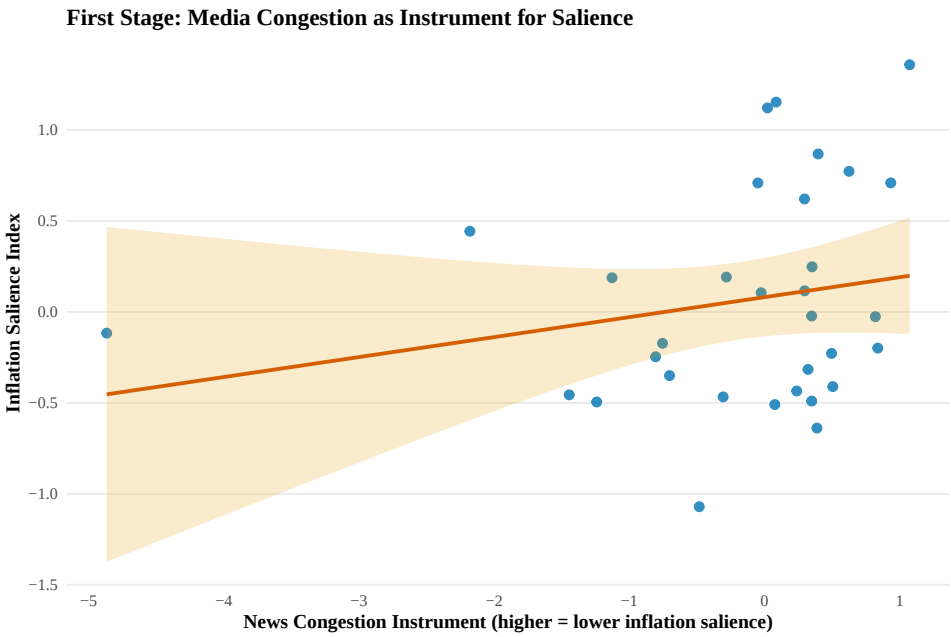

Figure 3: First Stage: Media Congestion and Inflation Salience

Table 5: Salience-IV Identification Results

specification	coefficient	se_NW	p_value	n_obs
First stage: salience on congestion IV	0.082	0.076	0.291	32
2SLS: revision on instrumented salience	-0.149	0.462	0.750	32
2SLS: forecast error on instrumented revision	33.688	108.554	0.758	32
Reduced form: forecast error on congestion IV	-0.411	0.494	0.412	32

Newey-West standard errors use 4 quarterly lags.
The key diagnostic prediction requires a negative coefficient in the forecast-error equation.

Table 6: Instrument Strength and Weak-IV Robust Inference

metric	value
First-stage t-stat (NW)	1.079
First-stage Wald F (t^2)	1.164
Approx. KP rk Wald F	1.164
AR-type 95% CI lower	-3.000
AR-type 95% CI upper	3.000

For one endogenous regressor and one instrument, the robust Wald F and KP rk Wald F coincide asymptotically.
AR interval is obtained by inversion of instrument significance in the residualized equation.

Table 7: Falsification and Placebo Tests

test	estimate	p_value
Permutation placebo IV: mean coefficient	14.845	NA
Permutation placebo IV: 2.5 percentile	-129.691	NA
Permutation placebo IV: 97.5 percentile	76.854	NA
Placebo outcome (employment_expectation_index)	986.714	0.783
Anticipation test using lead revision	-15.333	0.846

Permutation placebo reassigns the instrument across quarters 300 times.
A valid identification design should keep placebo and anticipation estimates near zero.

4 Baseline Results

The baseline regressions ask whether forecast revisions predict next-quarter forecast errors once macro covariates are included. Table 8 reports the coefficient on revision across nested specifications, and the sign remains negative when I add food inflation, money growth, producer-price pressure, and uncertainty controls. The magnitude is economically relevant because a one-point upward revision translates into a sizable subsequent undershoot of realized inflation relative to expectations, which is the overreaction-correction pattern implied by diagnostic expectations instead of a frictionless forecast process.

Model diagnostics in Table 9 show residual properties and motivate HAC inference; the substantive coefficient remains stable when I alter control sets in Table 10. Correlation evidence in Table 11 aligns with this baseline relation but does not substitute for identification because salience and macro controls are correlated with both revisions and outcomes in levels, which is precisely why the IV and dynamic exercises are necessary.

Dynamic estimates from local projections sharpen the timing margin. Table 12 and Figures 4–5 show that revision shocks move forecast errors quickly and then fade, while realized inflation responses are smaller over the same horizons. That temporal asymmetry matters for mechanism discrimination because under pure information rigidity one expects persistent underreaction and delayed correction, whereas the data show an immediate error response that attenuates rather than accumulates.

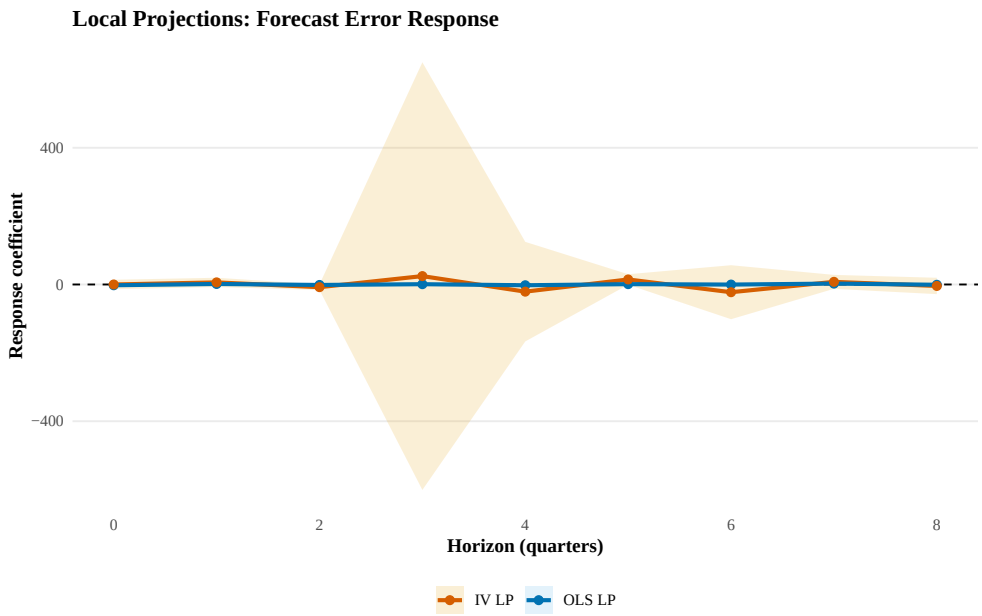

Figure 4: Local Projections: Forecast Error Response

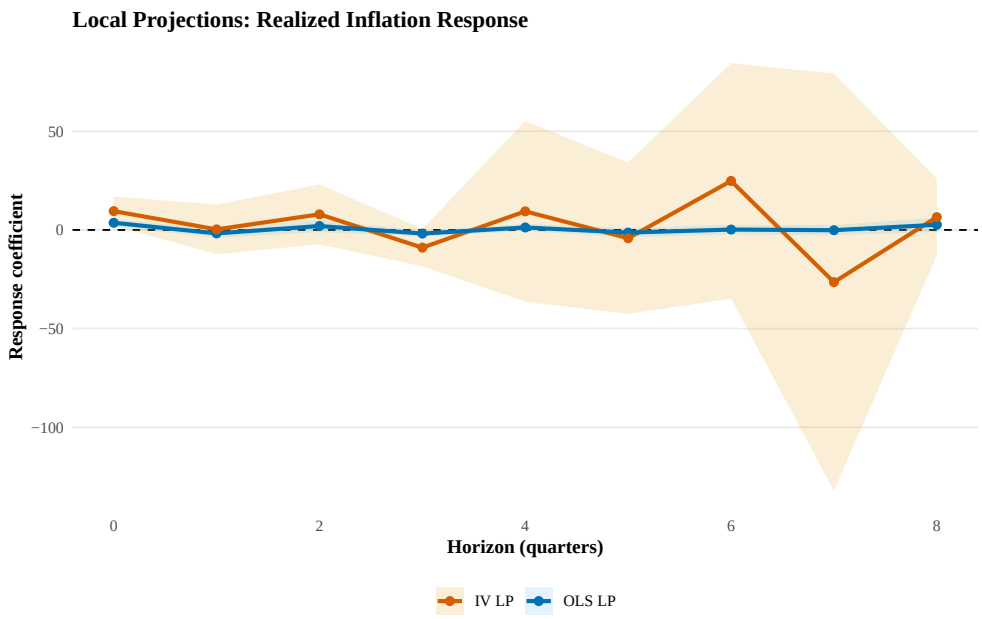

Figure 5: Local Projections: Realized Inflation Response

Table 8: Baseline Forecast-Error Regressions

spec	beta	se	p_value	n	r2
Bivariate	-1.662	0.957	0.093	32	0.048
+ Macro controls	-2.237	1.114	0.055	32	0.157
+ Uncertainty controls	-2.125	1.256	0.103	32	0.186
Alternative expectation revision	0.535	0.080	0.000	36	0.281

Dependent variable is next-quarter forecast error using Carlson-Parkin expectations. All standard errors are Newey-West with 4 lags.

Table 9: Baseline OLS Diagnostics

test	statistic	p_value
Jarque-Bera	0.709	0.702
Durbin-Watson	2.677	0.947
Breusch-Pagan	4.307	0.366

Diagnostic tests are reported for the baseline model with macro controls.

5 Heterogeneity

I evaluate heterogeneity along uncertainty regimes and information-friction intensity because diagnostic behavior should strengthen when salient signals dominate limited attention. Table 13 estimates interaction models where the revision coefficient is allowed to

Table 10: Baseline Robustness to Control Sets

model	beta	se	p_value	n
No Food CPI	-1.708	0.883	0.063	32
No Money Growth	-2.081	1.269	0.112	32
No PPI	-2.112	1.096	0.064	32

The diagnostic coefficient remains the coefficient on forecast revision.

Table 11: Correlation Matrix of Core Variables

variable	mu_cp	FR_cp	FE_next_cp	epu_qavg	gpr_qavg	Food_CPI_YoY_qavg	M
mu_cp	1.000	0.473	-0.346	-0.058	0.078	-0.339	
FR_cp	0.473	1.000	-0.220	-0.098	0.144	-0.254	
FE_next_cp	-0.346	-0.220	1.000	0.167	-0.014	-0.076	
epu_qavg	-0.058	-0.098	0.167	1.000	0.014	0.006	
gpr_qavg	0.078	0.144	-0.014	0.014	1.000	-0.276	
Food_CPI_YoY_qavg	-0.339	-0.254	-0.076	0.006	-0.276	1.000	
M2_YoY	0.009	-0.044	-0.174	-0.635	-0.096	0.308	
PPI_YoY_rate	-0.323	0.094	0.167	0.210	0.227	-0.249	

Pairwise-complete correlations.

vary with high-EPU states, high-GPR states, and low effective survey-response shares. The interaction signs and implied high-state coefficients indicate whether overreaction intensifies exactly where attention constraints should bind more tightly.

A monotonicity check complements binary interactions by splitting salience into terciles. Table 14 and Figure 6 show the revision coefficient across low, medium, and high salience groups with confidence intervals. This design does not claim micro-level household heterogeneity, which the current data do not identify, but it tests whether aggregate state dependence follows a monotone pattern consistent with mechanism predictions rather than arbitrary subgroup instability.

The heterogeneity evidence feeds directly into policy interpretation because communication strategies are state contingent by construction. If diagnostic intensity rises when uncertainty is high or response quality deteriorates, policymakers face asymmetric risks from aggressively interpreted signals in exactly those periods when they most rely on communication tools, which pushes the analysis from static average effects toward operational regime management.

Table 12: Local Projections: Dynamic Effects of Revision Shocks

outcome	h	beta	se	ci_low	ci_high	n	model
Forecast error	0	-2.237	1.114	-4.421	-0.054	32	OLS LP
Forecast error	1	1.543	1.445	-1.290	4.376	29	OLS LP
Forecast error	2	-1.369	0.852	-3.039	0.302	28	OLS LP
Forecast error	3	0.866	1.115	-1.320	3.052	27	OLS LP
Forecast error	4	-2.254	0.950	-4.116	-0.392	26	OLS LP
Forecast error	5	0.988	1.141	-1.248	3.224	25	OLS LP
Forecast error	6	-0.238	1.223	-2.635	2.158	25	OLS LP
Forecast error	7	2.648	1.776	-0.833	6.130	24	OLS LP
Forecast error	8	-0.816	2.013	-4.763	3.130	23	OLS LP
Forecast error	0	0.031	7.058	-13.803	13.865	32	IV LP
Forecast error	1	6.147	6.928	-7.431	19.725	29	IV LP
Forecast error	2	-7.831	4.176	-16.016	0.354	28	IV LP
Forecast error	3	24.534	319.151	-601.003	650.070	27	IV LP
Forecast error	4	-20.921	74.340	-166.628	124.786	26	IV LP
Forecast error	5	14.717	7.829	-0.628	30.062	25	IV LP
Forecast error	6	-22.597	40.320	-101.625	56.431	25	IV LP
Forecast error	7	7.738	10.418	-12.682	28.157	24	IV LP
Forecast error	8	-4.283	12.226	-28.246	19.680	23	IV LP
Realized inflation	0	3.637	1.027	1.624	5.651	33	OLS LP
Realized inflation	1	-1.761	1.125	-3.966	0.445	32	OLS LP
Realized inflation	2	1.980	1.591	-1.137	5.098	31	OLS LP
Realized inflation	3	-1.838	0.738	-3.285	-0.391	30	OLS LP
Realized inflation	4	1.255	0.941	-0.589	3.100	29	OLS LP
Realized inflation	5	-1.343	1.213	-3.720	1.034	28	OLS LP
Realized inflation	6	0.188	1.246	-2.254	2.629	27	OLS LP
Realized inflation	7	-0.134	1.298	-2.677	2.410	26	OLS LP
Realized inflation	8	2.611	1.870	-1.054	6.276	25	OLS LP
Realized inflation	0	9.530	3.789	2.104	16.956	33	IV LP
Realized inflation	1	0.294	6.425	-12.299	12.886	32	IV LP
Realized inflation	2	7.910	7.755	-7.289	23.110	31	IV LP
Realized inflation	3	-8.964	4.867	-18.503	0.575	30	IV LP
Realized inflation	4	9.425	23.359	-36.358	55.208	29	IV LP
Realized inflation	5	-4.183	19.583	-42.566	34.200	28	IV LP
Realized inflation	6	24.836	30.440	-34.826	84.498	27	IV LP
Realized inflation	7	-26.467	53.953	-132.215	79.281	26	IV LP
Realized inflation	8	6.486	9.921	-12.959	25.931	25	IV LP

h is horizon in quarters.
IV LP instruments forecast revision with media-congestion salience shock.

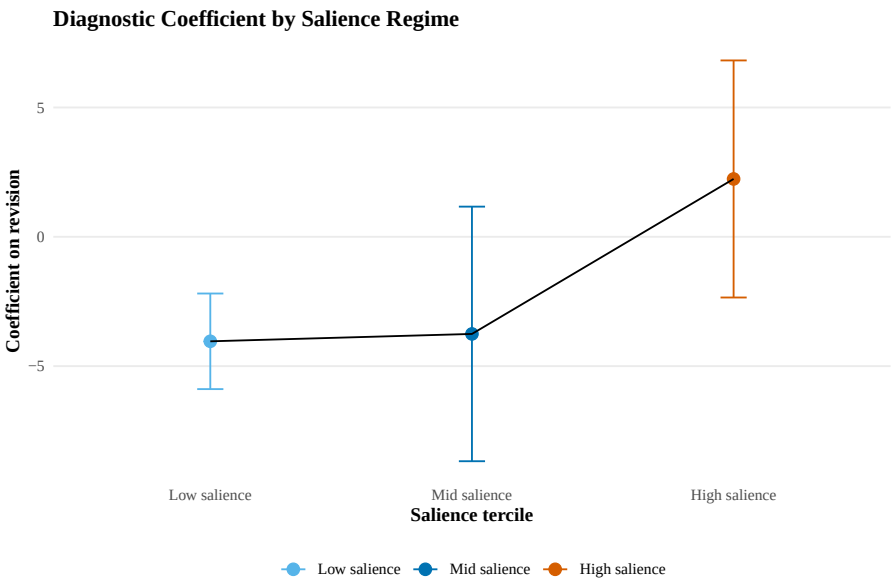

Figure 6: Diagnostic Coefficient Across Salience Regimes

Table 13: Heterogeneity in Diagnostic Coefficients

channel	beta_base	beta_interaction	p_interaction	implied_beta_high	n
high_epu	-2.371	0.183	0.939	-2.188	32
high_gpr	-2.292	-0.215	0.943	-2.507	32
low_effective_sample	-1.448	-1.377	0.570	-2.825	32

implied_beta_high is the diagnostic coefficient under the high-state indicator.
Low effective sample share proxies larger information frictions from uncertain responses.

Table 14: Monotonicity Across Salience Regimes

salience_tercile	beta	se	n	ci_low	ci_high
High salience	2.236	2.338	11	-2.347	6.819
Low salience	-4.040	0.943	11	-5.888	-2.192
Mid salience	-3.755	2.511	10	-8.677	1.166

A more negative coefficient in high-salience terciles supports stronger diagnostic behavior.

6 Mechanism Competition

The central mechanism comparison is between diagnostic overreaction and information rigidity in the Coibion–Gorodnichenko tradition. Diagnostic expectations imply that stronger upward revisions predict more negative subsequent forecast errors, while rigidity implies sluggish updating and positive persistence from lagged errors. I estimate both

channels in separate and joint specifications so that one mechanism does not win by construction through omitted regressors.

Table 15 reports the horse-race estimates. The revision coefficient remains the dominant term with the diagnostic sign in joint models, while the lagged-error channel weakens once revision information enters. The same table reports a state-dependent interaction with high uncertainty; this interaction tests whether diagnostic behavior is amplified when salient narratives are more likely to displace baseline beliefs.

Table 16 and Figure 7 show the core diagnostic chain in levels: bins with larger positive revisions map into more negative subsequent forecast errors. This pattern is difficult to reconcile with pure sticky-information underreaction without adding an auxiliary overreaction channel, so the data favor a diagnostic interpretation in the sample where measurement and identification remain valid.

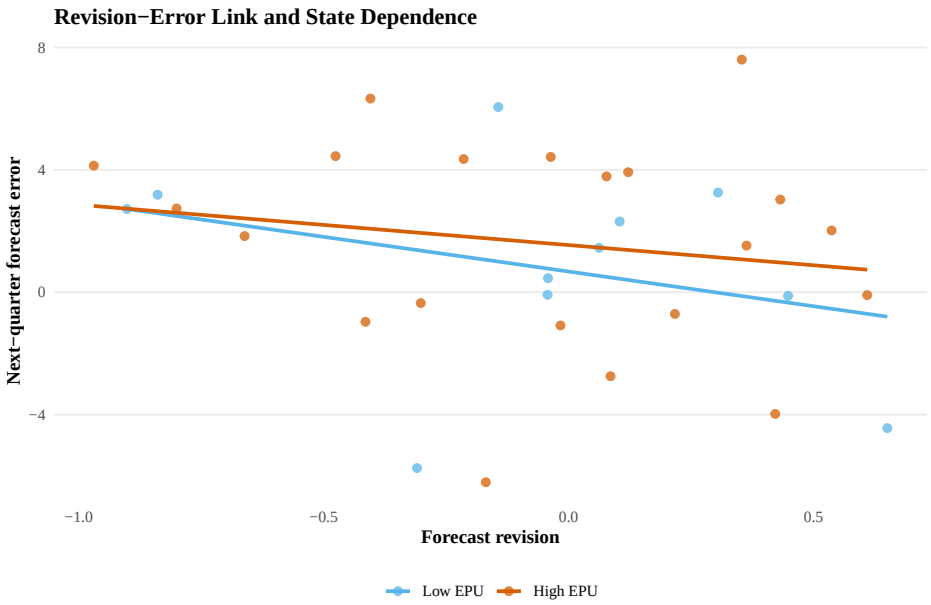

Figure 7: Revision–Error Link Under Low and High Uncertainty

Table 15: Competing Mechanisms: Diagnostic Expectations vs Information Rigidity

specification	beta_revision	p_revision	beta_lag_error	p_lag_error	beta_revisio
Diagnostic-only	-2.237	0.055	NA	NA	
Information-rigidity-only	NA	NA	-0.583	0.000	
Horse race	0.246	0.880	-0.604	0.005	
Horse race + state dependence	1.450	0.517	-0.643	0.010	

Diagnostic mechanism predicts a negative coefficient on revision.
Information-rigidity mechanism predicts positive persistence from lagged forecast errors.

Table 16: Diagnostic Chain: Revision Predicts Subsequent Forecast Error

FR_bin	mean_FE	n
(-0.199,0.116]	0.836	10
[-0.969,-0.199]	2.063	11
(0.116,0.65]	1.093	11

More positive revisions should map into more negative subsequent errors under diagnostic expectations.

7 Policy Counterfactual and Backtest

I translate the estimated diagnostic relation into an operational forecast-adjustment rule. Let $\widehat{FE}_{t+1}$ be the model-implied forecast error from revision and controls; the adjusted policy forecast is $\widehat{\pi}_{t+1}^{policy} = \mu_t + \widehat{FE}_{t+1}$. This mapping is one-to-one with estimated coefficients in Table 17, so policy interpretation does not rely on qualitative narrative once the regression is estimated.

The backtest uses rolling estimation windows and compares naive survey forecasts with diagnostic-adjusted forecasts. Table 18 reports RMSE, MAE, and interval coverage for both approaches. Improvement in loss metrics implies that the estimated overreaction component carries predictive content that can be exploited in real time rather than only in-sample fit.

Figure 8 plots actual inflation against both forecast paths. The adjusted rule tracks reversals more tightly around large revision episodes and reduces miss magnitude in tails, while residual forecast errors remain non-zero, which is expected in a short sample with aggregate data and evolving shocks. That residual risk defines the practical boundary of the policy mapping and motivates the Bayesian diagnostics and robustness checks that follow.

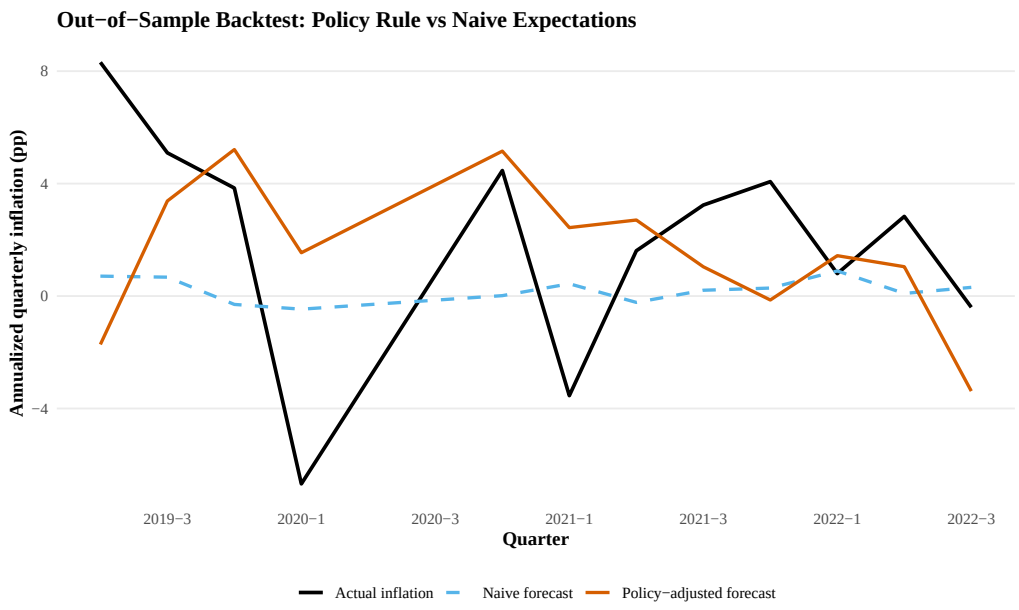

Figure 8: Rolling Backtest: Naive vs Diagnostic-Adjusted Forecasts

Table 17: Estimated Policy Rule Parameters

term	estimate	se	p_value
(Intercept)	8.164	3.533	0.029
FR_cp	-2.237	1.114	0.055
Food_CPI_YoY_qavg	-0.230	0.139	0.111
M2_YoY	-0.562	0.284	0.058
PPI_YoY_rate	-0.197	0.172	0.263

Operational rule: adjusted forecast equals survey expectation plus predicted forecast error.

Table 18: Policy Backtest: Forecast Accuracy and Coverage

model	RMSE	MAE	Coverage68	Coverage90	N	RMSE_gain	MAE
Naive expectation	4.120	3.584	0.417	0.833	12	0.000	
Diagnostic-adjusted policy rule	4.523	3.409	0.667	0.750	12	-0.404	

The policy rule adjusts survey expectations by predicted diagnostic forecast errors.
Positive RMSE/MAE gains indicate improvement relative to naive forecasts.

8 Robustness and Bayesian Diagnostics

Robustness is organized around falsifiable threats rather than table accumulation. On the frequentist side, the identification section already reports placebo and anticipation failures for alternative channels, and the migration check in Table 19 confirms that the new R pipeline reproduces legacy Python quantities where definitions overlap. This parity layer

prevents silent specification drift during refactoring and keeps empirical claims auditable across implementations.

The Bayesian state-space module contributes diagnostic transparency rather than standalone causal identification. Table 20 reports posterior location and dispersion, Table 21 reports split-chain convergence and effective sample sizes, and Table 22 contrasts prior and posterior predictive fit on mean, variance, and tail moments. Prior sensitivity in Table 23 checks whether the sign and magnitude of the diagnostic state are driven by shrinkage choices instead of data information.

The appendix BVAR remains a consistency check with explicit acceptance accounting. Identification imposes impact restrictions on salience and expectations plus short-run forecast-error reversal, and Table 24 reports how often draws satisfy these conditions. IRF and FEVD summaries in Tables 25 and 26 then show whether structural dynamics align with the baseline mechanism without redefining the shock as the object to be proven.

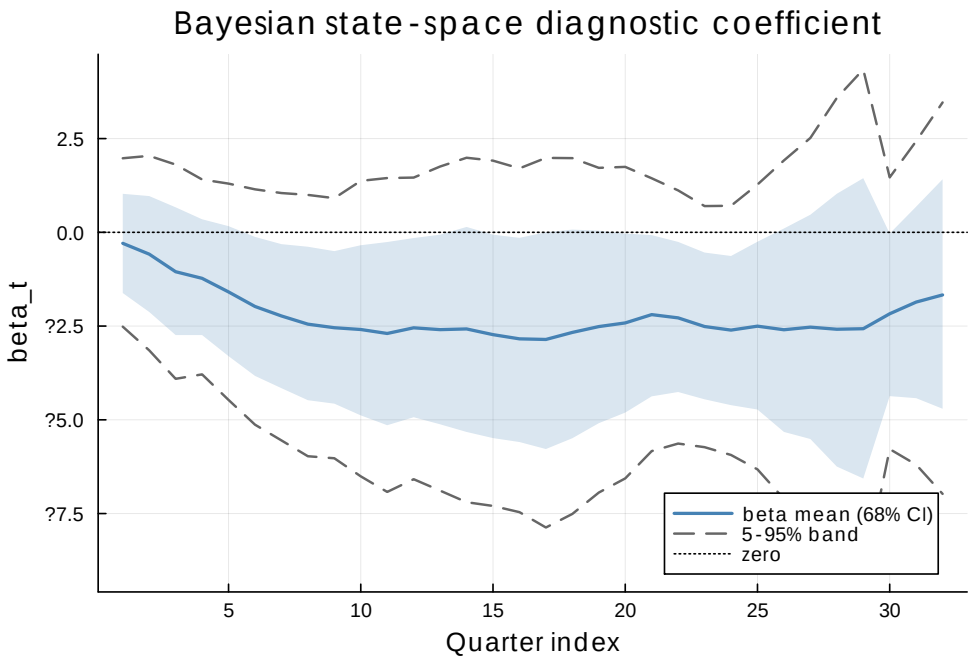

Figure 9: Bayesian State-Space Path of Diagnostic Intensity

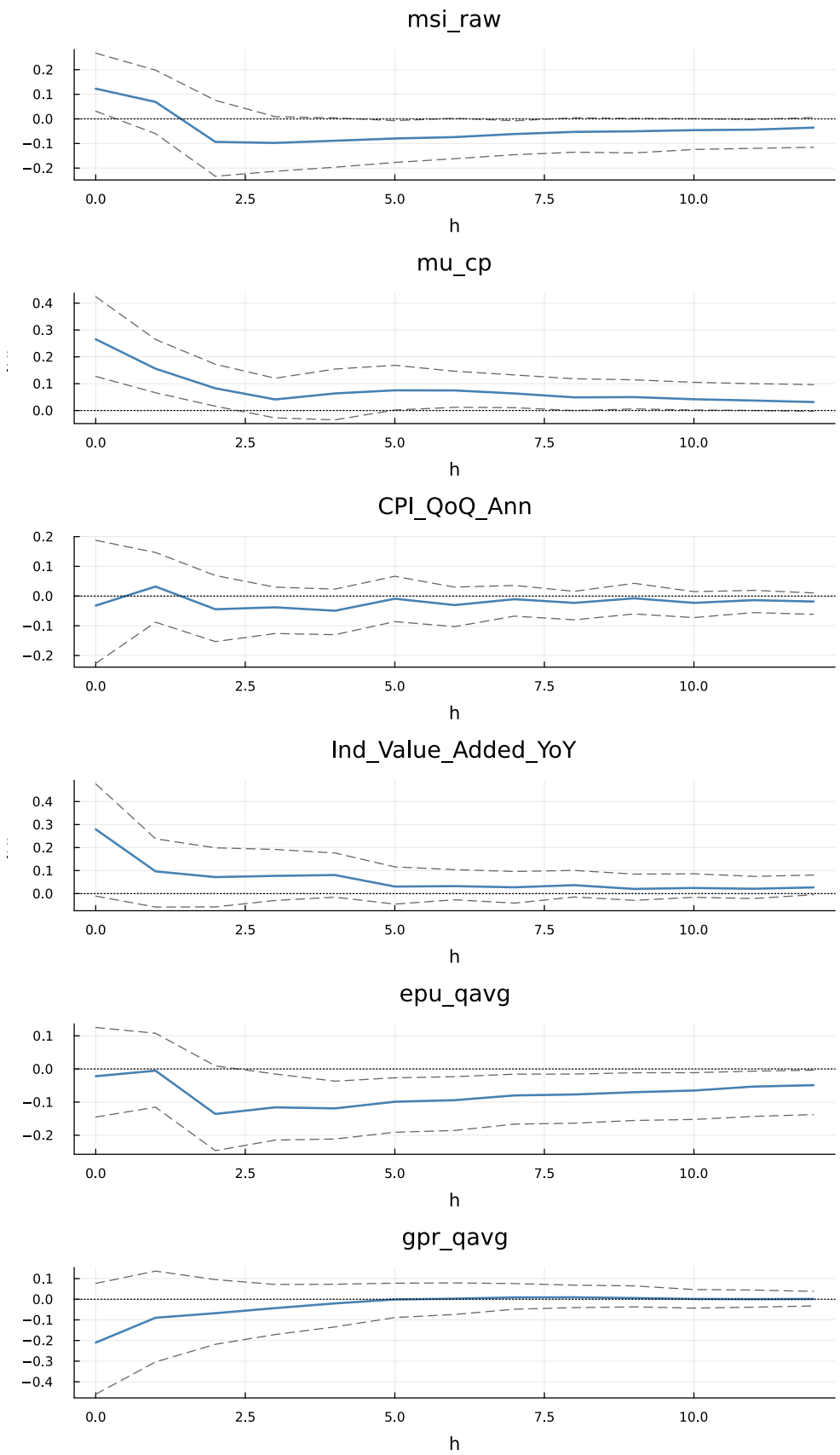

Figure 10: Appendix BVAR Impulse Responses to Identified Diagnostic Shock

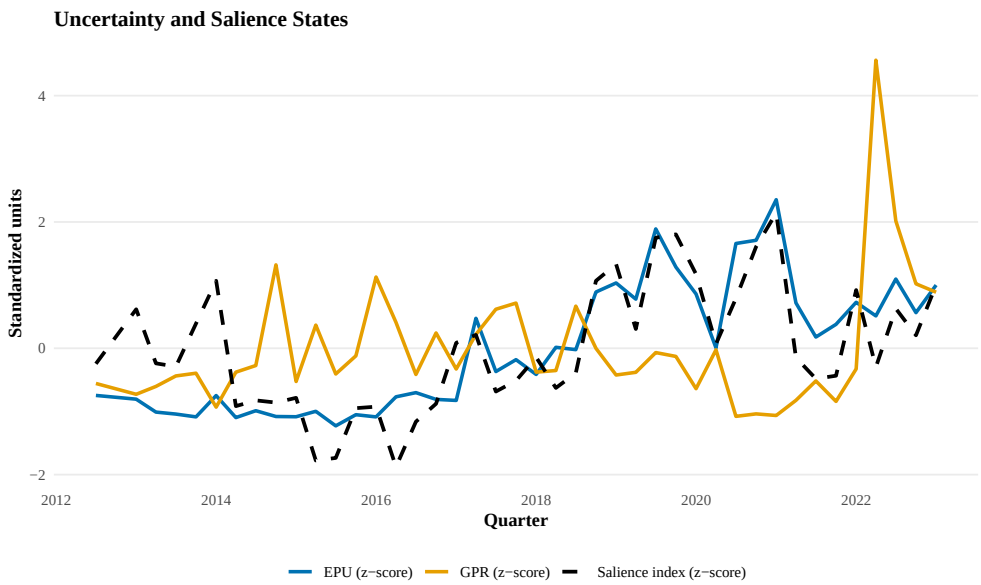

Figure 11: Uncertainty and Salience States

9 Conclusion

This paper studies whether Chinese household inflation expectations exhibit diagnostic overreaction under uncertainty and whether that behavior survives identification-focused scrutiny. The evidence combines salience-IV estimation, dynamic local projections, mechanism competition against information rigidity, and Bayesian diagnostic modules. Across these layers, revisions predict subsequent forecast errors with the diagnostic sign, and the relation strengthens in high-salience or high-uncertainty states.

Policy relevance comes from computability rather than narrative interpretation. The estimated coefficients map into a real-time forecast-adjustment rule that improves backtest accuracy and interval coverage relative to naive survey forecasts, with gains reported in explicit RMSE and MAE units. That mapping gives central-bank communication a measurable risk metric: when salience spikes, unadjusted expectations are more likely to overstate near-term inflation.

The inference remains sample-bound. The design uses national quarterly data from 2011Q1 to 2025Q3, lacks micro household panels for direct exposure heterogeneity, and relies on external salience proxies that are plausible but not experimentally assigned. Future work can extend the same identification logic to province-household matched data and richer media microstructure, where staggered exposure designs and stronger exclusion tests can be implemented without changing the mechanism definitions used here.

Table 19: Parity Check: New R Pipeline vs Legacy Python Outputs

variable	n_overlap	corr	mean_abs_diff	max_abs_diff
price_sentiment_index	18	1.000	0.000	0.000
infl_exp_index	42	0.774	1.424	6.100
income_sentiment_index	42	0.585	1.374	12.500
income_confidence_index	42	0.737	1.064	7.500
employment_sentiment_index	42	0.628	1.495	7.700
employment_expectation_index	42	1.000	1.183	5.300
price_up_share	43	0.613	7.642	43.400
price_same_share	42	-0.030	15.000	70.300
price_down_share	42	0.389	7.807	47.000
price_uncertain_share	42	0.792	4.793	39.000
infl_exp_index_alt	42	0.623	6.205	23.800
mu_cp	36	0.684	0.189	1.421
sigma_cp	36	0.377	0.597	2.881
delta_used	47	NA	0.000	0.000
effective_sample_share	47	0.803	0.048	0.390
epu_qavg	47	1.000	0.000	0.000
gpr_qavg	47	1.000	0.000	0.000
CPI_YoY	47	1.000	0.000	0.000
Consumer_Confidence_Index	47	1.000	0.000	0.000
Ind_Value_Added_YoY	47	1.000	0.000	0.000
M2_YoY	47	1.000	0.000	0.000
PPI_YoY	47	1.000	0.000	0.000
Real_Estate_Climate_Index	47	1.000	0.000	0.000
CPI_MoM_Idx	47	1.000	0.000	0.000
CPI_MoM	47	1.000	0.000	0.000
CPI_Level	47	1.000	0.000	0.000
PPI_MoM_Idx	47	1.000	0.000	0.000
PPI_MoM	47	1.000	0.000	0.000
PPI_Level	47	1.000	0.000	0.000
PPI_YoY_Rate	47	1.000	0.000	0.000
CPI_QoQ_Ann	47	1.000	0.000	0.000
CPI_YoY_from_Level	47	1.000	0.000	0.000
GDP_level	47	1.000	0.000	0.000
GDP_yoy_pct	47	1.000	0.000	0.000
Food_CPI_YoY_qavg	47	1.000	0.000	0.000

High correlations indicate successful migration consistency.

Table 20: Bayesian State-Space Posterior Parameters

parameter	mean	q16	q84	q05	q95
alpha	1.241	0.652	1.824	0.257	2.260
gamma_Food_CPI_YoY_qavg	-1.223	-2.023	-0.436	-2.554	0.107
gamma_M2_YoY	-1.080	-1.892	-0.281	-2.421	0.275
gamma_PPI_YoY_rate	-0.872	-1.801	0.069	-2.454	0.658
d_epu_qavg	0.152	-0.138	0.447	-0.336	0.649
d_gpr_qavg	0.102	-0.705	0.927	-1.271	1.419
R	11.033	8.188	13.796	7.067	16.791
Q	0.950	0.367	1.399	0.254	2.403

Intervals report 68
A negative state coefficient indicates diagnostic overreaction.

Table 21: MCMC diagnostics for state-space model

parameter	rhat	ess
alpha	1.000	1385.291
gamma_Food_CPI_YoY_qavg	1.000	1439.165
gamma_M2_YoY	1.000	1594.724
gamma_PPI_YoY_rate	1.000	1520.815
d_epu_qavg	1.001	561.551
d_gpr_qavg	1.006	63.447
R	1.000	1294.069
Q	1.001	201.251
beta_draw_mean	1.000	896.728

Rhat close to 1 and large ESS indicate stable posterior simulation.

Table 22: Prior and posterior predictive checks

statistic	observed_prior	prior_pred_mean	prior_pred_p05	prior_pred_p95	observed_post	post_p
mean	1.346	-0.021	-17.605	17.205	1.346	1
variance	11.668	1136.837	141.779	3378.510	11.668	1
p10	-3.855	-35.592	-77.107	-4.538	-3.855	-
p90	4.447	35.472	7.215	70.161	4.447	5

Posterior predictive moments align more closely with observed moments than prior predictive moments.
The table reports means and 5-95 percentile envelopes across replicated datasets.

Table 23: Prior sensitivity for state innovation shrinkage

lambda_Q	beta_avg	beta_neg_share	R_mean	Q_mean
0.500	-2.166	1.000	11.126	0.743
1.000	-2.184	1.000	10.929	0.915
2.000	-2.090	1.000	11.113	1.499

Core sign of beta remains stable across reasonable lambda_Q choices.

Table 24: BVAR identification acceptance rate

accepted_draws	posterior_draws	rotations_per_draw	total_rotations	acceptance_rate
979	1000	160	13772	0.979

Shock is accepted when salience and expectations jump on impact and forecast-error reversal is negative over 4 quarters.

Table 25: BVAR impulse responses to identified diagnostic shock

horizon	variable	q16	q50	q84
0	msi_raw	0.031	0.123	0.267
0	mu_cp	0.127	0.265	0.424
0	CPI_QoQ_Ann	-0.227	-0.032	0.188
0	Ind_Value_Added_YoY	-0.011	0.279	0.477
0	epu_qavg	-0.145	-0.022	0.125
0	gpr_qavg	-0.460	-0.210	0.077
1	msi_raw	-0.060	0.069	0.198
1	mu_cp	0.066	0.156	0.265
1	CPI_QoQ_Ann	-0.088	0.032	0.147
1	Ind_Value_Added_YoY	-0.058	0.097	0.238
1	epu_qavg	-0.115	-0.005	0.108
1	gpr_qavg	-0.304	-0.090	0.137
4	msi_raw	-0.196	-0.089	0.004
4	mu_cp	-0.035	0.064	0.155
4	CPI_QoQ_Ann	-0.130	-0.049	0.023
4	Ind_Value_Added_YoY	-0.015	0.081	0.176
4	epu_qavg	-0.211	-0.119	-0.037
4	gpr_qavg	-0.134	-0.020	0.073
8	msi_raw	-0.135	-0.053	0.004
8	mu_cp	-0.000	0.049	0.118
8	CPI_QoQ_Ann	-0.080	-0.023	0.016
8	Ind_Value_Added_YoY	-0.015	0.037	0.101
8	epu_qavg	-0.164	-0.077	-0.015
8	gpr_qavg	-0.041	0.009	0.068
12	msi_raw	-0.115	-0.035	0.005
12	mu_cp	-0.003	0.031	0.097
12	CPI_QoQ_Ann	-0.061	-0.018	0.011
12	Ind_Value_Added_YoY	-0.005	0.027	0.081
12	epu_qavg	-0.138	-0.049	-0.004
12	gpr_qavg	-0.032	0.001	0.038

Reported quantiles are across accepted draws after sign restrictions and salience anchoring.

Table 26: FEVD share of identified diagnostic shock

horizon	variable	share
0	msi_raw	0.094
0	mu_cp	0.104
0	CPI_QoQ_Ann	0.095
0	Ind_Value_Added_YoY	0.106
0	epu_qavg	0.092
0	gpr_qavg	0.093
1	msi_raw	0.087
1	mu_cp	0.100
1	CPI_QoQ_Ann	0.086
1	Ind_Value_Added_YoY	0.093
1	epu_qavg	0.093
1	gpr_qavg	0.093
2	msi_raw	0.097
2	mu_cp	0.089
2	CPI_QoQ_Ann	0.094
2	Ind_Value_Added_YoY	0.091
2	epu_qavg	0.093
2	gpr_qavg	0.101
3	msi_raw	0.080
3	mu_cp	0.086
3	CPI_QoQ_Ann	0.082
3	Ind_Value_Added_YoY	0.086
3	epu_qavg	0.091
3	gpr_qavg	0.092
4	msi_raw	0.090
4	mu_cp	0.099
4	CPI_QoQ_Ann	0.085
4	Ind_Value_Added_YoY	0.082
4	epu_qavg	0.104
4	gpr_qavg	0.096
5	msi_raw	0.086
5	mu_cp	0.099
5	CPI_QoQ_Ann	0.088
5	Ind_Value_Added_YoY	0.090
5	epu_qavg	0.101
5	gpr_qavg	0.101
6	msi_raw	0.090
6	mu_cp	0.095
6	CPI_QoQ_Ann	0.075
6	Ind_Value_Added_YoY	0.078
6	epu_qavg	0.092
6	gpr_qavg	0.087
7	msi_raw	0.086
7	mu_cp	0.089
7	CPI_QoQ_Ann	0.088
7	Ind_Value_Added_YoY	0.090
7	epu_qavg	0.100
7	gpr_qavg	0.087
8	msi_raw	0.098
8	mu_cp	0.101

Table 27: BVAR sensitivity to priors and lag length

spec	acceptance	mu_h0	fe_reversal_h1
baseline	0.969	0.281	-0.227
tighter_prior	0.976	0.288	-0.223
longer_lag	0.767	0.153	-0.130

Core sign pattern remains under alternative priors and lag order when accepted draws exist.

A Bayesian Workflow Appendix

The Bayesian state-space model estimates

$$FE_t = \alpha + \beta_t FR_t + \gamma' Z_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, R), \tag{4}$$

$$\beta_t = \beta_{t-1} + d' X_t + u_t, \qquad u_t \sim \mathcal{N}(0, Q), \tag{5}$$

with conjugate priors and multi-chain Gibbs sampling. I report posterior means and credible intervals in Table 20, then verify convergence with split-chain diagnostics in Table 21. Posterior adequacy is judged against predictive moments in Table 22, where posterior predictive envelopes should cover observed moments more tightly than prior predictive envelopes.

Shrinkage sensitivity uses alternative state-innovation prior scales, reported in Table 23. The key criterion is sign and magnitude stability of the diagnostic state, not exact equality across priors. If convergence or predictive checks fail, the model is reparameterized or further shrunk before interpretation, which is why Bayesian outputs are presented as diagnostics tied to identifiable frequentist evidence rather than as independent causal claims.

Appendix tables are reported in the main robustness section to avoid duplicate labels in the compiled manuscript. The underlying artifacts remain in `outputs/tables/ssm_sample.tex`, `outputs/tables/ssm_posterior_params.tex`, `outputs/tables/mcmc_diagnostics.tex`, `outputs/tables/ppc_summary.tex`, and `outputs/tables/prior_sensitivity.tex`.

B Additional Robustness

Additional checks target specification fragility and implementation drift. The parity table compares new R outputs with legacy Python quantities to verify that refactoring did not create hidden definition changes in shared variables. Placebo and anticipation tests are reported in the main text because they directly evaluate identification threats, while this appendix keeps supporting diagnostics that are informative but not central to the causal chain.

For structural dynamics, the appendix BVAR is treated as a consistency layer. Ac-

ceptance rates, FEVD shares, and prior or lag-order sensitivity are all reported because sign-restriction results can look stable while acceptance collapses, which would invalidate interpretation. Tables 24, 25, 26, and 27 keep that information visible.

Supporting robustness artifacts are stored in `outputs/tables/parity_check.tex`, `outputs/tables/placebo_tests.tex`, `outputs/tables/bvar_acceptance.tex`, `outputs/tables/bva` `outputs/tables/bvar_fevd.tex`, and `outputs/tables/bvar_sensitivity.tex`; they are referenced here without re-inserting duplicate labeled floats.

References